

**Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember***

Laporan Akhir Praktikum Jaringan Komputer

VLAN Trunk OSPF Multivendor

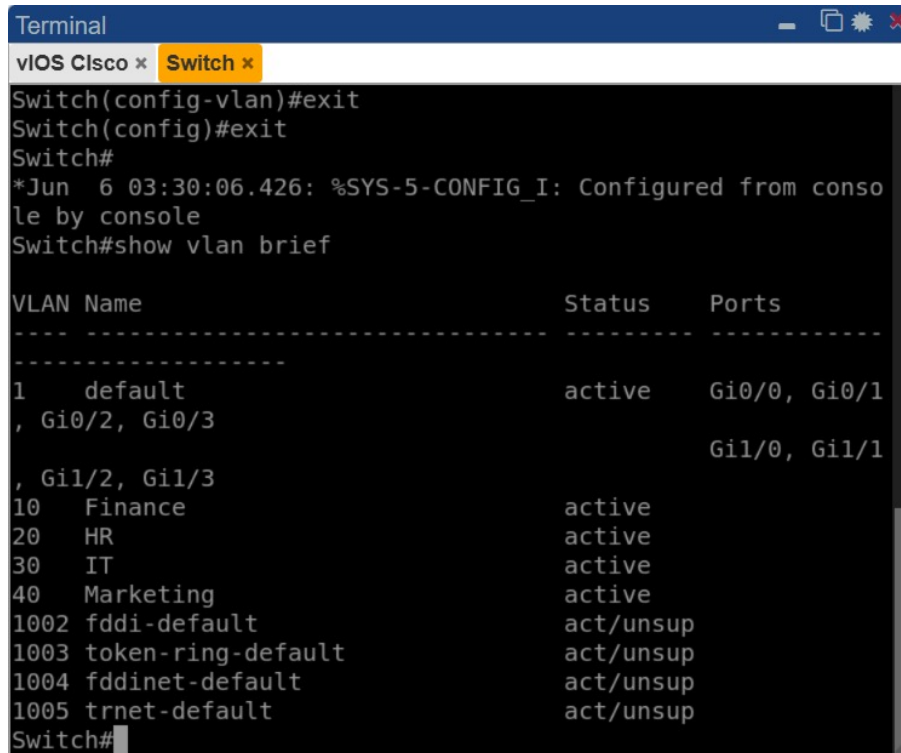
Verdia Nachel Tunggal Dewi - 5024241012

2026

1 Langkah-Langkah Percobaan

1.1 Konfigurasi Cisco Switch

1. Masuk ke terminal Cisco Switch
2. Buat VLAN, lalu verifikasi apakah statusnya sudah aktif

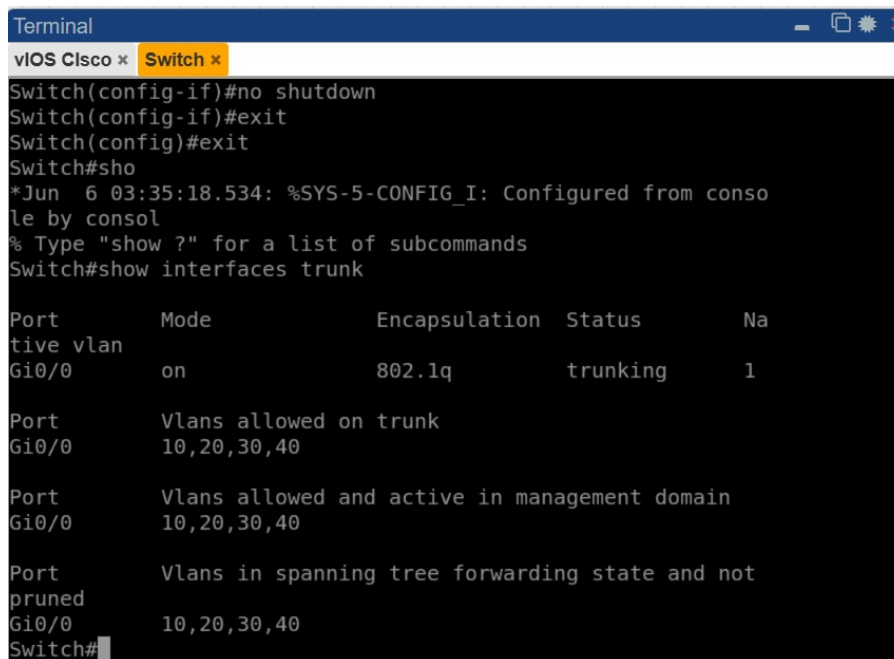


```
Terminal
vIOS Cisco x Switch x
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
*Jun  6 03:30:06.426: %SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Gi0/0, Gi0/1
, Gi0/2, Gi0/3
, Gi1/0, Gi1/1
, Gi1/2, Gi1/3
10   Finance                 active
20   HR                     active
30   IT                     active
40   Marketing               active
1002 fddi-default            act/unsup
1003 token-ring-default     act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup
Switch#
```

Gambar 1: Verifikasi VLAN

3. Konfigurasi access port untuk setiap VLAN
4. Konfigurasi Trunk ke Cisco Router, lalu verifikasi dan save semua perintah



```
Terminal
vIOS Cisco x Switch x
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#exit
Switch#sho
*Jun  6 03:35:18.534: %SYS-5-CONFIG_I: Configured from console by console
% Type "show ?" for a list of subcommands
Switch#show interfaces trunk

Port        Mode           Encapsulation  Status        Native vlan
Gi0/0       on             802.1q         trunking      1

Port        Vlans allowed on trunk
Gi0/0       10,20,30,40

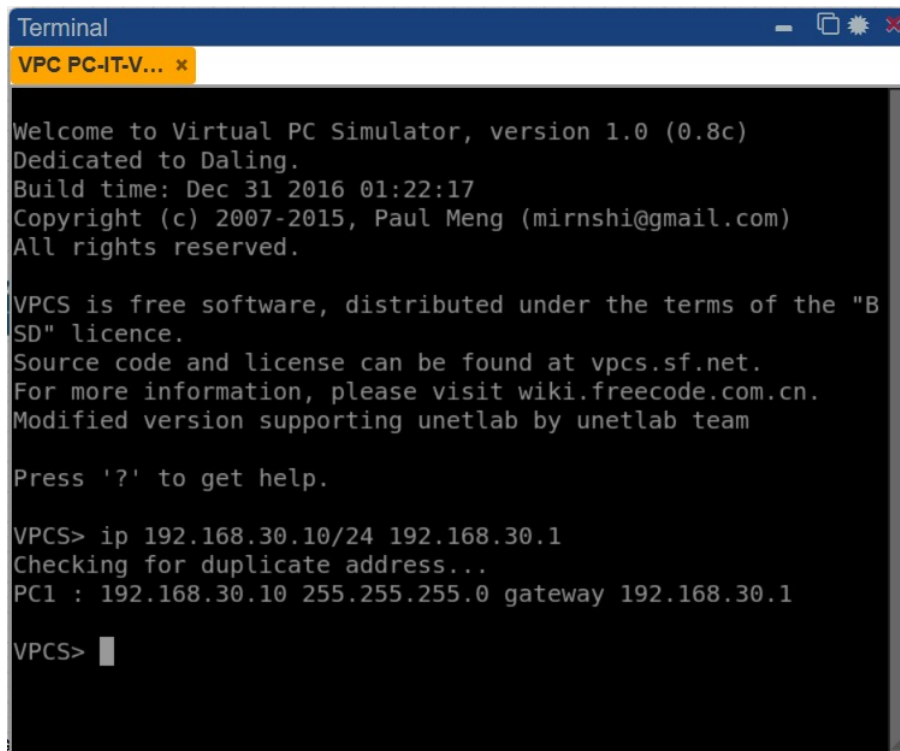
Port        Vlans allowed and active in management domain
Gi0/0       10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Gi0/0       10,20,30,40
Switch#
```

Gambar 2: Verifikasi trunk port

1.2 Konfigurasi Cisco Router

1. Masuk ke Cisco Router
2. Konfigurasi interface trunk ke switch
3. Konfigurasi Subinterface tiap VLAN
4. Konfigurasi DHCP-Client untuk tiap VLAN
5. Konfigurasi IP static pada VPCS untuk VLAN 30 dan 40, dan DHCP untuk VLAN 10 dan 20



```
Terminal
VPC PC-IT-V... x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

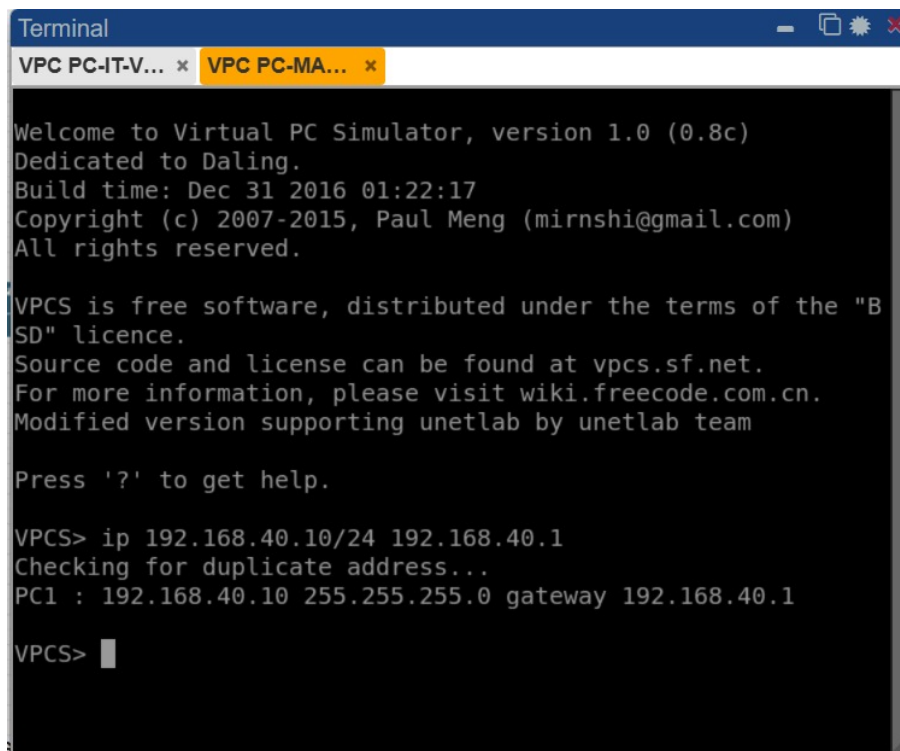
VPCS is free software, distributed under the terms of the "B
SD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS> █
```

Gambar 3: Konfigurasi IP static pada PC VLAN 30 (IT)



```
Terminal
VPC PC-IT-V... x VPC PC-MA... x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "B
SD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
PC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS> █
```

Gambar 4: Konfigurasi IP static pada PC VLAN 40 (Marketing)

```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x

Press '?' to get help.

VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> ip dhcp
DORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86397, 86400/43200/75600
MAC        : 00:50:79:66:68:9b
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> 
```

Gambar 5: PC VLAN 10 mendapatkan IP via DHCP

```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x

Press '?' to get help.

VPCS> ip dhcp
DDORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> ip dhcp
DORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 86395, 86400/43200/75600
MAC        : 00:50:79:66:68:9a
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> 
```

Gambar 6: PC VLAN 20 mendapatkan IP via DHCP

6. Verifikasi Cisco Router

```

Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
*Jun  6 03:49:22.982: %SYS-5-CONFIG_I: Configured from console
le by console
% Type "show ?" for a list of subcommands
Router#show ip interface brief
Interface                IP-Address      OK? Method Status
      Protocol
GigabitEthernet0/0      unassigned      YES unset  admini
stratively down down
GigabitEthernet0/1      unassigned      YES unset  admini
stratively down down
GigabitEthernet0/2      unassigned      YES unset  up
up
GigabitEthernet0/2.10    192.168.10.1    YES manual  up
up
GigabitEthernet0/2.20    192.168.20.1    YES manual  up
up
GigabitEthernet0/2.30    192.168.30.1    YES manual  up
up
GigabitEthernet0/2.40    192.168.40.1    YES manual  up
up
GigabitEthernet0/3      unassigned      YES unset  admini
stratively down down
Router#

```

Gambar 7: Verifikasi VLAN 10, 20, 30, 40

```

Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
Pool VLAN10-FINANCE :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index      IP address range      Leased ad
dresses
192.168.10.12      192.168.10.1 - 192.168.10.254  1
Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index      IP address range      Leased ad
dresses
192.168.20.12      192.168.20.1 - 192.168.20.254  1
Router#

```

Gambar 8: Verifikasi pool VLAN 10 dan VLAN 20

7. Uji jaringan melalui VPCS dan verifikasi IP DHCP binding

```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86397, 86400/43200/75600
MAC        : 00:50:79:66:68:9b
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.135 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=2.610 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.629 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.602 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.202 ms

VPCS>
```

Gambar 9: PC VLAN 10 ping gateway 192.168.10.1

```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 86395, 86400/43200/75600
MAC        : 00:50:79:66:68:9a
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=2.904 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.744 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.960 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=3.726 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=3.562 ms

VPCS>
```

Gambar 10: PC VLAN 20 ping gateway 192.168.20.1

```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
DHCP LEASE : 86395, 86400/43200/75600
MAC : 00:50:79:66:68:9a
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500

VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=2.904 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.744 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.960 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=3.726 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=3.562 ms

VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=3.084 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=3.336 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=2.880 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=3.303 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=3.262 ms

VPCS> 
```

Gambar 11: PC VLAN 20 ping gateway VLAN 20 dan VLAN 30

```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=2.904 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.744 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.960 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=3.726 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=3.562 ms

VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=3.084 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=3.336 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=2.880 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=3.303 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=3.262 ms

VPCS> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=3.172 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=3.483 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=3.394 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=3.611 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=4.493 ms

VPCS> 
```

Gambar 12: PC VLAN 20 ping gateway VLAN 30 dan VLAN 40


```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
DHCP LEASE : 86397, 86400/43200/75600
MAC : 00:50:79:66:68:9b
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500

VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.135 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=2.610 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.629 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.602 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.202 ms

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=10.095 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=5.725 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=4.266 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=5.099 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=5.338 ms

VPCS> 
```

Gambar 13: PC VLAN 10 ping gateway dan ping ke VLAN 20

```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=10.095 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=5.725 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=4.266 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=5.099 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=5.338 ms

VPCS> ping 192.168.30.10

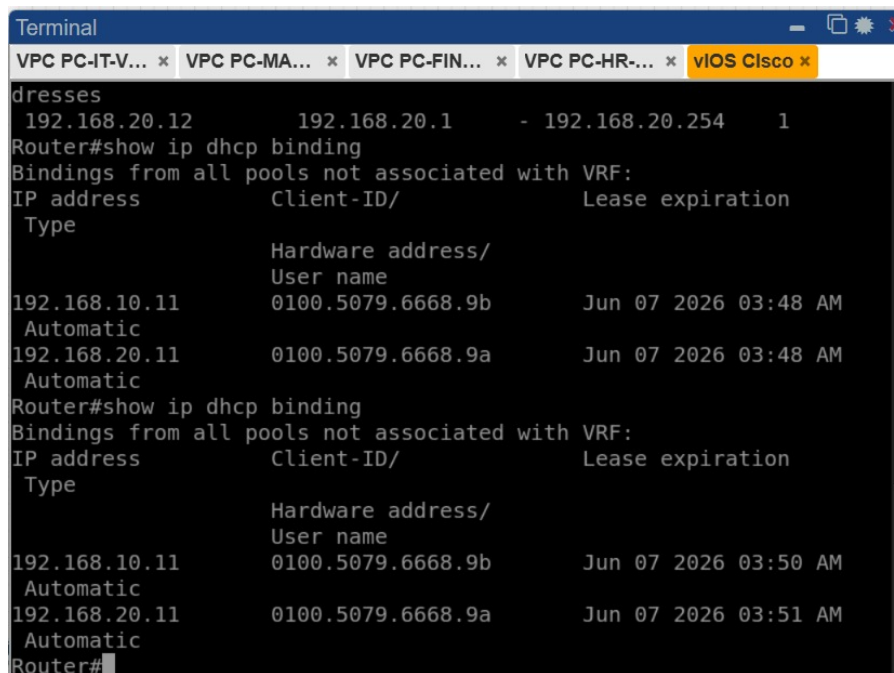
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=11.953 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=5.356 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=6.404 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=5.350 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=4.446 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=12.243 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=5.812 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=5.189 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=5.945 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=6.182 ms

VPCS> 
```

Gambar 14: PC VLAN 10 ping ke VLAN 30 dan VLAN 40

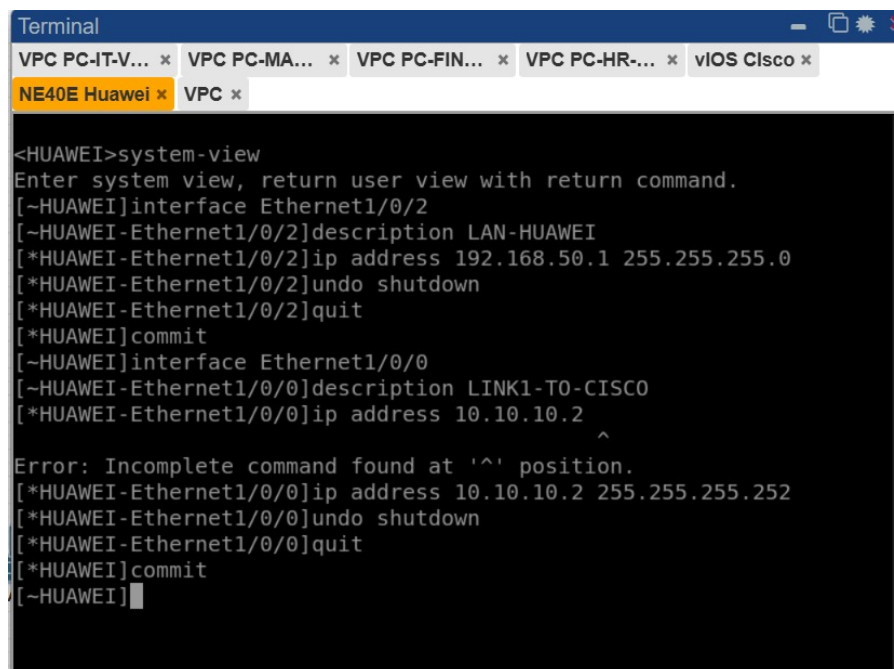


```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x VIOS Cisco x
dresses
192.168.20.12      192.168.20.1      - 192.168.20.254    1
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration
Type
Hardware address/
User name
192.168.10.11    0100.5079.6668.9b      Jun 07 2026 03:48 AM
Automatic
192.168.20.11    0100.5079.6668.9a      Jun 07 2026 03:48 AM
Automatic
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration
Type
Hardware address/
User name
192.168.10.11    0100.5079.6668.9b      Jun 07 2026 03:50 AM
Automatic
192.168.20.11    0100.5079.6668.9a      Jun 07 2026 03:51 AM
Automatic
Router#
```

Gambar 15: Verifikasi dhcp binding

1.3 Konfigurasi Huawei Router dan IP Address Antarrouter

1. Konfigurasi LAN Huawei



```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x VIOS Cisco x
NE40E Huawei x VPC x
<HUAWEI>system-view
Enter system view, return user view with return command.
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2]description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2]ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2]undo shutdown
[*HUAWEI-Ethernet1/0/2]quit
[*HUAWEI]commit
[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2
Error: Incomplete command found at '^' position.
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0]undo shutdown
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]commit
[~HUAWEI]
```

Gambar 16: Konfigurasi interface LAN Huawei

2. Konfigurasi IP VPC LAN Huawei

```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
NE40E Huawei x VPC x
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 192.168.50.10/24
GATEWAY     : 192.168.50.1
DNS         :
MAC         : 00:50:79:66:68:9c
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

VPCS> 
```

Gambar 17: Konfigurasi IP static pada PC LAN Huawei

3. Uji PING gateway LAN Huawei

```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
NE40E Huawei x VPC x

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 192.168.50.10/24
GATEWAY     : 192.168.50.1
DNS         :
MAC         : 00:50:79:66:68:9c
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=10.206 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.545 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=3.943 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=3.062 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=3.134 ms

VPCS> 
```

Gambar 18: PC LAN Huawei ping gateway

4. Konfigurasi IP link Cisco ↔ Huawei link 1

```

Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
NE40E Huawei x VPC x

<HUAWEI>system-view
Enter system view, return user view with return command.
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2]description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2]ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2]undo shutdown
[*HUAWEI-Ethernet1/0/2]quit
[*HUAWEI]commit
[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2
Error: Incomplete command found at '^' position.
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0]undo shutdown
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]commit
[~HUAWEI]

```

Gambar 19: Konfigurasi Ethernet1/0/0 Huawei

5. Konfigurasi IP link Cisco ↔ Huawei link 2

```

Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
NE40E Huawei x VPC x

ce GigabitEthernet0/3, changed state to up
Router(config)#exit
Router#
*Jun 6 04:02:30.581: %SYS-5-CONFIG_I: Configured from console by console
Router#interface gi0/0
Router(config-if)#description LINK2-T0-HUAWEI
Router(config-if)#ip address 10.10.11.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 6 04:05:55.712: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 6 04:05:56.713: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

```

Gambar 20: Konfigurasi interface Gi0/0 Cisco Router

6. Konfigurasi IP link Huawei ↔ Mikrotik

7. Konfigurasi IP link Cisco ↔ Mikrotik


```

Terminal
Mikrotik x vIOS Cisco x

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    10.10.11.1      YES manual up          up
GigabitEthernet0/1    10.10.30.1      YES manual up          up
GigabitEthernet0/2    unassigned      YES unset  up          up
GigabitEthernet0/2.10 192.168.10.1    YES manual up          up
GigabitEthernet0/2.20 192.168.20.1    YES manual up          up
GigabitEthernet0/2.30 192.168.30.1    YES manual up          up
GigabitEthernet0/2.40 192.168.40.1    YES manual up          up
GigabitEthernet0/3    10.10.10.1      YES manual up          up
Router#

```

Gambar 23: Verifikasi pada Cisco Router

```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x

[~HUAWEI]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface      IP Address/Mask  Physical  Protocol  VPN
Ethernet1/0/0    10.10.10.2/30    up        up        --
Ethernet1/0/0.4094 128.1.138.137/16 up        up        l3vpn
Ethernet1/0/1    unassigned       up        down      --
Ethernet1/0/1.4094 128.1.138.137/16 up        up        l3vpn
Ethernet1/0/2    192.168.50.1/24  up        up        --
Ethernet1/0/2.4094 128.1.138.137/16 up        up        l3vpn
Ethernet1/0/3    10.10.20.1/30    up        up        --
Ethernet1/0/3.4094 128.1.138.137/16 up        up        l3vpn
Ethernet1/0/4    10.10.11.2/30    up        up        --
Ethernet1/0/4.4094 128.1.138.137/16 up        up        l3vpn
---- More ----

```

Gambar 24: Verifikasi pada Huawei Router


```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ether2 comment=TO
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ether2 comment=T
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ether2 comment=
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ether2 comment=t
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ether2 comment=
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ether2 comment
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ether2 comme
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ether2 comm
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ether2 com
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ether2 co
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ether2 c
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ether2
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ether2
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ether
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ethe
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=eth
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=eh
[admin@MikroTik] > ip address add address=10.10.30.2/30 interface=ehe
[admin@MikroTik] > ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;; TO-HUAWEI
10.10.20.2/30 10.10.20.0 ether1
1 ;; TO-CISCO
10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] >

```

Gambar 25: Verifikasi pada Mikrotik

9. Uji jaringan antarrouter

```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.11.1 YES manual up up
GigabitEthernet0/1 10.10.30.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset up up
GigabitEthernet0/2.10 192.168.10.1 YES manual up up
GigabitEthernet0/2.20 192.168.20.1 YES manual up up
GigabitEthernet0/2.30 192.168.30.1 YES manual up up
GigabitEthernet0/2.40 192.168.40.1 YES manual up up
GigabitEthernet0/3 10.10.10.1 YES manual up up
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/8 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/6 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/2 ms
Router#

```

Gambar 26: Cisco Router ping ke Huawei Link 1, Link 2, dan MikroTik

```
Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x
[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.10.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms
```

Gambar 27: Huawei Router ping ke Cisco Link 1 dan Link 2

```
Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
  Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=2 ms
  Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

--- 10.10.20.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]
```

Gambar 28: Huawei Router ping ke MikroTik


```
Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
[admin@MikroTik] > ping 10.10.20.1
  SEQ HOST                                SIZE TTL TIME  ST..
    0 10.10.20.1                          56 255 3ms
    1 10.10.20.1                          56 255 1ms
    2 10.10.20.1                          56 255 1ms
    3 10.10.20.1                          56 255 2ms
    4 10.10.20.1                          56 255 1ms
    5 10.10.20.1                          56 255 1ms
    6 10.10.20.1                          56 255 1ms
    7 10.10.20.1                          56 255 5ms
    8 10.10.20.1                          56 255 1ms
    9 10.10.20.1                          56 255 1ms
   10 10.10.20.1                          56 255 1ms
  sent=11 received=11 packet-loss=0% min-rtt=1ms avg-rtt=1ms
  max-rtt=5ms

[admin@MikroTik] > ping 10.10.30.1
  SEQ HOST                                SIZE TTL TIME  ST..
    0 10.10.30.1                          56 255 1ms
    1 10.10.30.1                          56 255 1ms
    2 10.10.30.1                          56 255 1ms
    3 10.10.30.1                          56 255 1ms
    4 10.10.30.1                          56 255 1ms
    5 10.10.30.1                          56 255 1ms
  sent=6 received=6 packet-loss=0% min-rtt=1ms avg-rtt=1ms
  max-rtt=1ms

[admin@MikroTik] > 
```

Gambar 29: MikroTik ping ke Huawei dan Cisco

1.4 Konfigurasi OSPF Multivendor dan OSPF Cost

1. Konfigurasi OSPF dan verifikasi pada Cisco Router

```
Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/8 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/6 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/2 ms
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 10.10.10.0
% Incomplete command.

Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 10.10.11.0 0.0.0.3 area 0
Router(config-router)#network 10.10.30.0 0.0.0.3 area 0
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#
```

Gambar 30: Konfigurasi OSPF Cisco Router

```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
Router(config)#interface gi0/0
Router(config-if)#description LINK2-TO-HUAWEI
Router(config-if)#ip ospf cost 10
Router(config-if)#exit
Router(config)#interface gi0/1
Router(config-if)#description LINK-TO-MIKROTIK
Router(config-if)#ip ospf cost 100
Router(config-if)#exit
Router(config)#exit
Router#show
*Jun  6 04:27:14.789: %SYS-5-CONFIG_I: Configured from console by console ip
% Incomplete command.

Router#show ip ospf neighbor
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```

Gambar 31: Konfigurasi OSPF cost pada interface Cisco dan verifikasi

2. Verifikasi route pada Cisco

```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance         Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance         Last Update
  Distance: (default is 110)

```

Gambar 32: Verifikasi route pada Cisco

3. Konfigurasi OSPF pada Huawei Router

```

Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x
[*HUAWEI]interface Ethernet1/0/3
[*HUAWEI-Ethernet1/0/3]description LINK2-TO-MIKROTIK
[*HUAWEI-Ethernet1/0/3]ospf cost 10
[*HUAWEI-Ethernet1/0/3]ospf cost 100
[*HUAWEI-Ethernet1/0/3]quit
[*HUAWEI]commit
[*HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
  State: Full          Mode:Nbr is Slave    Priority: 1
  DR: 10.10.10.1      BDR: 10.10.10.2      MTU: 1500
  Dead timer due in 34 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m11s
  Neighbor Up Time : 2026-06-06 04:33:45
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
  State: Full          Mode:Nbr is Slave    Priority: 1
  DR: 10.10.11.1      BDR: 10.10.11.2      MTU: 1500
  Dead timer due in 36 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m11s
  Neighbor Up Time : 2026-06-06 04:33:45
  Authentication Sequence: [ 0 ]

[*HUAWEI]

```

Gambar 33: Konfigurasi OSPF cost pada Huawei dan verifikasi

4. Verifikasi route OSPF pada Huawei

```

Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2
[*HUAWEI]display ospf routing

      OSPF Process 1 with Router ID 2.2.2.2
      Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
10.10.10.0/30    10        Direct    10.10.10.2    2.2.2.2        0.0.0.0
10.10.11.0/30    10        Direct    10.10.11.2    2.2.2.2        0.0.0.0
10.10.20.0/30    100       Direct    10.10.20.1    2.2.2.2        0.0.0.0
10.10.30.0/30    110       Stub      10.10.11.1    1.1.1.1        0.0.0.0
10.10.30.0/30    110       Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.50.0/24  1         Direct    192.168.50.1  2.2.2.2        0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

      OSPF Process 65534 with Router ID 128.1.138.137
      Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
128.1.138.137/32  0         Direct    128.1.138.137 128.1.138.137 0.0.0.0

Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0

```

Gambar 34: Verifikasi route pada Huawei

5. Konfigurasi OSPF pada Mikrotik

```

Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x
1 10.10.20.2/30 10.10.20.0 ether1
1 ;; TO-CISCO
10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] > ping 10.10.20.1
  SEQ HOST                                SIZE TTL TIME  ST..
  0 10.10.20.1                            56 255 3ms
  1 10.10.20.1                            56 255 1ms
  2 10.10.20.1                            56 255 1ms
  3 10.10.20.1                            56 255 2ms
  4 10.10.20.1                            56 255 1ms
  5 10.10.20.1                            56 255 1ms
  6 10.10.20.1                            56 255 1ms
  7 10.10.20.1                            56 255 5ms
  8 10.10.20.1                            56 255 1ms
  9 10.10.20.1                            56 255 1ms
 10 10.10.20.1                            56 255 1ms
    sent=11 received=11 packet-loss=0% min-rtt=1ms avg-rtt=1ms
    max-rtt=5ms
[admin@MikroTik] > /routing ospf netoTik] > routi
ng ospf network add network=10.10.3work add network=10.10.30.0/30 area=backbone
rea=backboneoTik] > routing ospf network add network=10.10.20.0/30 area=backbone
radmin@MikroTik] > routing ospf network add network=10.10.20.0/30 area=backbone
redmin@MikroTik] > routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=backbone
failure: configuration for this network already exists.10.30.0/30 area=backbone
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=backbon
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=backbo
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=backb
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=back
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=bac
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=ba
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=b
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=b
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area
[admin@MikroTik] >

```

Gambar 35: Konfigurasi OSPF network dan interface cost pada MikroTik

6. Verifikasi Mikrotik, pastikan router id Cisco dan Huawei sudah muncul di Mikrotik

```

Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x
5 10.10.20.1                            56 255 1ms
6 10.10.20.1                            56 255 1ms
7 10.10.20.1                            56 255 5ms
8 10.10.20.1                            56 255 1ms
9 10.10.20.1                            56 255 1ms
10 10.10.20.1                           56 255 1ms
    sent=11 received=11 packet-loss=0% min-rtt=1ms avg-rtt=1ms
    max-rtt=5ms
[admin@MikroTik] > /routing ospf netoTik] > routi
ng ospf network add network=10.10.3work add network=10.10.30.0/30 area=backbone
rea=backboneoTik] > routing ospf network add network=10.10.20.0/30 area=backbone
radmin@MikroTik] > routing ospf network add network=10.10.20.0/30 area=backbone
redmin@MikroTik] > routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=backbone
failure: configuration for this network already exists.10.30.0/30 area=backbone
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=backbon
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=backbo
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=backb
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=back
[admin@MikroTik] > routing ospf network add netin@MikroTik] > routing min@MikroTik] > routi
ng ospf interface add interface=ether2ace add interface=ether2 cost=100
=100min@MikroTik] > routing ospf interface add interface=ether1 cost=100
=100min@MikroTik] > routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1
  interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=23s
1 instance=default router-id=2.2.2.2 address=10.10.20.1
  interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=10.10.20.2 state="Full" state-changes=5

```

Gambar 36: Verifikasi OSPF neighbor Mikrotik dan routing OSPF

```

Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x
[admin@MikroTik] > routing ospf network add network=10.10.30.0/30 area=back
[admin@MikroTik] > routing ospf network add netin@MikroTik] > routing min@MikroTik] > routi
ng ospf interface add interface=ether2ace add interface=ether2 cost=100
=100min@MikroTik] > routing ospf interface add interface=ether1 cost=100
=admin@MikroTik] > routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1
  interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=23s

1 instance=default router-id=2.2.2.2 address=10.10.20.1
  interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=10.10.20.2 state="Full" state-changes=5
[admin@MikroTik] > routing ospf interface add interface=ether2 cost=100
failure: there already exists configuration for this interface cost=100
[admin@MikroTik] > ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#   DST-ADDRESS      PREF-SRC  GATEWAY
0 ADo 10.10.10.0/30
  10.10.20.1
  10.10.30.1
1 ADo 10.10.11.0/30
  10.10.20.1
  10.10.30.1
2 ADo 192.168.10.0/24
  10.10.30.1
3 ADo 192.168.20.0/24
  10.10.30.1
4 ADo 192.168.30.0/24
  10.10.30.1
5 ADo 192.168.40.0/24
  10.10.30.1
6 ADo 192.168.50.0/24
  10.10.20.1
[admin@MikroTik] >

```

Gambar 37: Verifikasi routing table Mikrotik

7. Verifikasi OSPF neighbor

```

Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x

Router con0 is now available

Press RETURN to get started.

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* education. IOSv is provided as-is and is not supported by Cisco's *
* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IOSv Software or Documentation to any third party for any *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****

Router>enable
Router#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
3.3.3.3        1     FULL/BDR        00:00:31   10.10.30.2    GigabitEthernet0/1
2.2.2.2        1     FULL/BDR        00:00:39   10.10.11.2    GigabitEthernet0/0
2.2.2.2        1     FULL/BDR        00:00:30   10.10.10.2    GigabitEthernet0/3
Router#

```

Gambar 38: Verifikasi OSPF neighbor pada Cisco

```
Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
  State: Full           Mode:Nbr is Slave   Priority: 1
  DR: 10.10.10.1       BDR: 10.10.10.2     MTU: 1500
  Dead timer due in 36 sec
  Retrans timer interval: 5
  Neighbor is up for 00h06m55s
  Neighbor Up Time : 2026-06-06 04:33:45
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3      Address: 10.10.20.2
  State: Full           Mode:Nbr is Master  Priority: 1
  DR: 10.10.20.1       BDR: 10.10.20.2     MTU: 1500
  Dead timer due in 39 sec
  Retrans timer interval: 5
  Neighbor is up for 00h02m19s
  Neighbor Up Time : 2026-06-06 04:38:21
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
  State: Full           Mode:Nbr is Slave   Priority: 1
  DR: 10.10.11.1       BDR: 10.10.11.2     MTU: 1500
  Dead timer due in 34 sec
  Retrans timer interval: 5
  Neighbor is up for 00h06m55s
```

Gambar 39: Verifikasi OSPF neighbor pada Huawei

```
terminal
Mikrotik x vIOS Cisco x NE40E Huawei x

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
  State: Full           Mode:Nbr is Slave   Priority: 1
  DR: 10.10.10.1       BDR: 10.10.10.2     MTU: 1500
  Dead timer due in 36 sec
  Retrans timer interval: 5
  Neighbor is up for 00h06m55s
  Neighbor Up Time : 2026-06-06 04:33:45
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3      Address: 10.10.20.2
  State: Full           Mode:Nbr is Master  Priority: 1
  DR: 10.10.20.1       BDR: 10.10.20.2     MTU: 1500
  Dead timer due in 39 sec
  Retrans timer interval: 5
  Neighbor is up for 00h02m19s
  Neighbor Up Time : 2026-06-06 04:38:21
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
  State: Full           Mode:Nbr is Slave   Priority: 1
  DR: 10.10.11.1       BDR: 10.10.11.2     MTU: 1500
  Dead timer due in 34 sec
  Retrans timer interval: 5
  Neighbor is up for 00h06m55s
  Neighbor Up Time : 2026-06-06 04:33:45
  Authentication Sequence: [ 0 ]
```

Gambar 40: Detail OSPF neighbor Huawei


```

Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x
0 instance=default router-id=1.1.1.1 address=10.10.30.1
  interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=23s

1 instance=default router-id=2.2.2.2 address=10.10.20.1
  interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=10.10.20.2 state="Full" state-changes=5
[admin@MikroTik] > routing ospf interface add interface=ether2 cost=100
failure: there already exists configuration for this interface cost=100
[admin@MikroTik] > ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#   DST-ADDRESS      PREF-SRC  GATEWAY
0 Ado 10.10.10.0/30      10.10.20.1
1 Ado 10.10.11.0/30      10.10.30.1
2 Ado 192.168.10.0/24     10.10.20.1
3 Ado 192.168.20.0/24     10.10.30.1
4 Ado 192.168.30.0/24     10.10.30.1
5 Ado 192.168.40.0/24     10.10.30.1
6 Ado 192.168.50.0/24     10.10.20.1
[admin@MikroTik] > routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1
  interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=3m25s

1 instance=default router-id=2.2.2.2 address=10.10.20.1
  interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=3m25s
[admin@MikroTik] >

```

Gambar 41: Verifikasi OSPF neighbor pada MikroTik

8. Verifikasi Routing Table

```

Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x

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* of the IOSv Software or Documentation to any third party for any *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing.
*****
Router>enable
Router#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
3.3.3.3        1     FULL/BDR        00:00:31    10.10.30.2     GigabitEthernet0/1
2.2.2.2        1     FULL/BDR        00:00:39    10.10.11.2     GigabitEthernet0/0
2.2.2.2        1     FULL/BDR        00:00:30    10.10.10.2     GigabitEthernet0/3

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:08:22 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:08:22 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:08:22 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1

```

Gambar 42: Cek route menuju Huawei pada Cisco


```

Router>enable
Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
3.3.3.3          1    FULL/BDR        00:00:31    10.10.30.2     GigabitEthernet0/1
2.2.2.2          1    FULL/BDR        00:00:39    10.10.11.2     GigabitEthernet0/0
2.2.2.2          1    FULL/BDR        00:00:30    10.10.10.2     GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:08:22 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:08:22 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:08:22 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O       10.10.20.0/30 [110/110] via 10.10.11.2, 00:08:51, GigabitEthernet0/0
         [110/110] via 10.10.10.2, 00:08:51, GigabitEthernet0/3
O       192.168.50.0/24 [110/11] via 10.10.11.2, 00:08:51, GigabitEthernet0/0
         [110/11] via 10.10.10.2, 00:08:51, GigabitEthernet0/3
Router#

```

Gambar 43: Verifikasi route OSPF pada Cisco

```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
^
Error: Unrecognized command found at '^' position.
[-HUAWEI]
[-HUAWEI]
[-HUAWEI]
[-HUAWEI]
[-HUAWEI]
[-HUAWEI]
[-HUAWEI]
[-HUAWEI]
[-HUAWEI]
[-HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
_public_ Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto    Pre    Cost    Flags NextHop          Interface
-----
10.10.30.0/30       OSPF     10     110     D    10.10.11.1    Ethernet1/0/4
10.10.10.0/24       OSPF     10     110     D    10.10.10.1    Ethernet1/0/0
192.168.10.0/24     OSPF     10     11      D    10.10.11.1    Ethernet1/0/4
192.168.20.0/24     OSPF     10     11      D    10.10.10.1    Ethernet1/0/0
192.168.30.0/24     OSPF     10     11      D    10.10.11.1    Ethernet1/0/4
192.168.40.0/24     OSPF     10     11      D    10.10.10.1    Ethernet1/0/0
192.168.50.0/24     OSPF     10     11      D    10.10.11.1    Ethernet1/0/4

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4
---- More ----

```

Gambar 44: Verifikasi route OSPF pada Huawei (1)

```

Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x
[~HUAWEI]
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto  Pre  Cost      Flags NextHop      Interface
10.10.30.0/30      OSPF   10   110       D   10.10.11.1      Ethernet1/0/4
                  OSPF   10   110       D   10.10.10.1      Ethernet1/0/0
192.168.10.0/24     OSPF   10   11         D   10.10.11.1      Ethernet1/0/4
                  OSPF   10   11         D   10.10.10.1      Ethernet1/0/0
192.168.20.0/24     OSPF   10   11         D   10.10.11.1      Ethernet1/0/4
                  OSPF   10   11         D   10.10.10.1      Ethernet1/0/0
192.168.30.0/24     OSPF   10   11         D   10.10.11.1      Ethernet1/0/4
                  OSPF   10   11         D   10.10.10.1      Ethernet1/0/0
192.168.40.0/24     OSPF   10   11         D   10.10.11.1      Ethernet1/0/4
                  OSPF   10   11         D   10.10.10.1      Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask    Proto  Pre  Cost      Flags NextHop      Interface
10.10.10.0/30       OSPF   10   10         10.10.10.2      Ethernet1/0/0
10.10.11.0/30       OSPF   10   10         10.10.11.2      Ethernet1/0/4
10.10.20.0/30       OSPF   10   100        10.10.20.1      Ethernet1/0/3
192.168.50.0/24     OSPF   10   1          192.168.50.1    Ethernet1/0/2

[~HUAWEI]
[~HUAWEI]
[~HUAWEI]

```

Gambar 45: Verifikasi route OSPF pada Huawei (2)

```

Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x
# DST-ADDRESS      PREF-SRC  GATEWAY
0 Ado 10.10.10.0/30    10.10.20.1
      10.10.30.1
1 Ado 10.10.11.0/30    10.10.20.1
      10.10.30.1
2 Ado 192.168.10.0/24   10.10.30.1
3 Ado 192.168.20.0/24   10.10.30.1
4 Ado 192.168.30.0/24   10.10.30.1
5 Ado 192.168.40.0/24   10.10.30.1
6 Ado 192.168.50.0/24 10.10.20.1

[admin@MikroTik] > routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1
  interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=3m25s
1 instance=default router-id=2.2.2.2 address=10.10.20.1
  interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=3m25s

[admin@MikroTik] > ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC  GATEWAY
0 Ado 10.10.10.0/30    10.10.20.1
      10.10.30.1
1 Ado 10.10.11.0/30    10.10.20.1
      10.10.30.1
2 Ado 192.168.10.0/24 10.10.30.1
3 Ado 192.168.20.0/24 10.10.30.1
4 Ado 192.168.30.0/24 10.10.30.1
5 Ado 192.168.40.0/24 10.10.30.1
6 Ado 192.168.50.0/24 10.10.20.1

[admin@MikroTik] >

```

Gambar 46: Verifikasi route OSPF pada MikroTik

9. Verifikasi koneksi end-to-end

```
Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x VPC PC-FIN... x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=10.585 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.004 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.077 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=5.717 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=7.570 ms
VPCS>
```

Gambar 47: PC VLAN 10 (Finance) ping ke PC LAN Huawei

```
Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.908 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.662 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=9.470 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.718 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=7.124 ms
VPCS>
```

Gambar 48: PC VLAN 20 (HR) ping ke PC LAN Huawei

```
Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-IT-V... x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.035 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.530 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.647 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.097 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.885 ms
VPCS>
```

Gambar 49: PC VLAN 30 (IT) ping ke PC LAN Huawei

```
Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-IT-V... x VPC PC-MA... x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.845 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.951 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.017 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=8.636 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=6.069 ms
VPCS>
```

Gambar 50: PC VLAN 40 (Marketing) ping ke PC LAN Huawei

```
Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-IT-V... x VPC PC-MA... x VPC x

VPCS> ping 192.168.10.11

84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=7.750 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=7.276 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=6.034 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=7.772 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=18.920 ms

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=7.928 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=11.389 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=8.516 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=6.475 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=5.919 ms

VPCS> ping 192.168.30.11

192.168.30.11 icmp_seq=1 timeout
192.168.30.11 icmp_seq=2 timeout
192.168.30.11 icmp_seq=3 timeout
192.168.30.11 icmp_seq=4 timeout
192.168.30.11 icmp_seq=5 timeout

VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=5.812 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=9.049 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=7.474 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=6.173 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=9.131 ms

VPCS>
```

Gambar 51: PC LAN Huawei ping ke VLAN 10, 20, 30, dan 40

1.5 Pengujian Failover OSPF

1. Kondisi normal: dua link Cisco-Huawei aktif

```
VPCS>
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    4.972 ms  3.289 ms  3.573 ms
 2  10.10.11.2     8.839 ms  4.800 ms  5.527 ms
 3  *192.168.50.10 11.816 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```

Gambar 52: Trace dari PC VLAN 10 ke LAN Huawei

2. Kondisi satu link Cisco-Huawei mati

```

Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-IT-V... x VPC PC-MA... x VPC x
0 192.168.50.0/24 [110/11] via 10.10.11.2, 00:08:51, GigabitEthernet0/0
[110/11] via 10.10.10.2, 00:08:51, GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:15:08 ago
Routing Descriptor Blocks:
10.10.11.2, from 2.2.2.2, 00:15:08 ago, via GigabitEthernet0/0
Route metric is 11, traffic share count is 1
* 10.10.10.2, from 2.2.2.2, 00:15:08 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)#shutdown
Router(config-if)#exi
*Jun 6 04:50:15.186: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from FULL to DOWN, Neighbor Down: Interface down or detached
Router(config)#
*Jun 6 04:50:17.154: %LINK-5-CHANGED: Interface GigabitEthernet0/3, changed state to administratively down
*Jun 6 04:50:18.154: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to down
Router(config)#exit
Router#
*Jun 6 04:50:23.296: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:17:15 ago
Routing Descriptor Blocks:
* 10.10.11.2, from 2.2.2.2, 00:17:15 ago, via GigabitEthernet0/0
Route metric is 11, traffic share count is 1
Router#

```

Gambar 53: Shutdown Gi0/3 dan cek route pada Cisco

```

VPCS> trace 192.168.50.10
1 192.168.50.10 11.016 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
1 192.168.10.1 3.751 ms 4.534 ms 5.273 ms
2 10.10.11.2 7.071 ms 3.725 ms 4.792 ms
3 *192.168.50.10 8.447 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>

```

Gambar 54: Trace dari PC VLAN ke LAN Huawei

3. Kondisi dua link Cisco-Huawei mati

```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-IT-V... x VPC PC-MA... x VPC x
Router(config)#exit
Router#
*Jun 6 04:50:23.296: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:17:15 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:17:15 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/0
Router(config-if)#shutdown
Router(config-if)#
*Jun 6 04:51:42.084: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
*Jun 6 04:51:44.053: %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down
*Jun 6 04:51:45.053: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
Router(config-if)#exit
Router(config)#exit
Router#show i
*Jun 6 04:51:53.608: %SYS-5-CONFIG_I: Configured from console by console
% Incomplete command.

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:00:18 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:00:18 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
Router#

```

Gambar 55: Shutdown Gi0/0 dan cek route pada Cisco

```

3 *192.168.50.10 8.447 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.236 ms 3.592 ms 3.744 ms
 2 10.10.30.2 7.805 ms 3.025 ms 4.226 ms
 3 10.10.20.1 8.634 ms 3.781 ms 7.546 ms
 4 *192.168.50.10 10.856 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>

```

Gambar 56: Trace dari PC VLAN ke LAN Huawei

4. Aktifkan kembali dua link Cisco-Huawei

```
Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-IT-V... x VPC PC-MA... x VPC x
Router(config)#interface gi0/0
*Jun 6 04:52:58.468: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOA
DING to FULL, Loading Done
Router(config-if)#interface gi0/3
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 6 04:53:12.395: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to up
*Jun 6 04:53:13.395: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, c
hanged state to upe
% Ambiguous command: "e"
Router(config)#
Router(config)#
Router(config)#
Router(config)#exit
Router#
*Jun 6 04:53:17.892: %SYS-5-CONFIG I: Configured from console by consolehow
% Type "show ?" for a list of subcommands
Router#
Router#
*Jun 6 04:53:22.320: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOA
DING to FULL, Loading Donesho
% Type "show ?" for a list of subcommands
Router#
Router#
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.10.2 on GigabitEthernet0/3, 00:00:10 ago
Routing Descriptor Blocks:
* 10.10.11.2, from 2.2.2.2, 00:00:30 ago, via GigabitEthernet0/0
Route metric is 11, traffic share count is 1
10.10.10.2, from 2.2.2.2, 00:00:10 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
Router#
```

Gambar 57: Aktifkan Gi0/3 dan Gi0/0 dan cek route pada cisco

```
Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-IT-V... x VPC PC-MA... x VPC x
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=10.585 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.004 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.077 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=5.717 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=7.570 ms

VPCS>
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 4.972 ms 3.289 ms 3.573 ms
 2 10.10.11.2 8.839 ms 4.800 ms 5.527 ms
 3 *192.168.50.10 11.816 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.751 ms 4.534 ms 5.273 ms
 2 10.10.11.2 7.071 ms 3.725 ms 4.792 ms
 3 *192.168.50.10 8.447 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.236 ms 3.592 ms 3.744 ms
 2 10.10.30.2 7.805 ms 3.025 ms 4.226 ms
 3 10.10.20.1 8.634 ms 3.781 ms 7.546 ms
 4 *192.168.50.10 10.856 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 11.861 ms 5.401 ms 4.251 ms
 2 10.10.11.2 8.124 ms 6.930 ms 4.013 ms
 3 *192.168.50.10 11.307 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```

Gambar 58: Trace dari PC VLAN ke LAN Huawei

2 Analisis Hasil Percobaan

Pada Praktikum 1 dan 2, Cisco Switch berhasil dikonfigurasi dengan VLAN 10 (Finance), 20 (HR), 30 (IT), dan 40 (Marketing), serta trunk port menuju Cisco Router. Cisco Router dikonfigurasi menggunakan metode Router-on-a-Stick (ROAS) dengan subinterface untuk setiap VLAN, sekaligus berfungsi

si sebagai DHCP Server untuk VLAN 10 dan 20. Hasil pengujian menunjukkan seluruh PC berhasil mendapatkan IP sesuai VLAN-nya dan dapat melakukan ping antar-VLAN melalui Cisco Router.

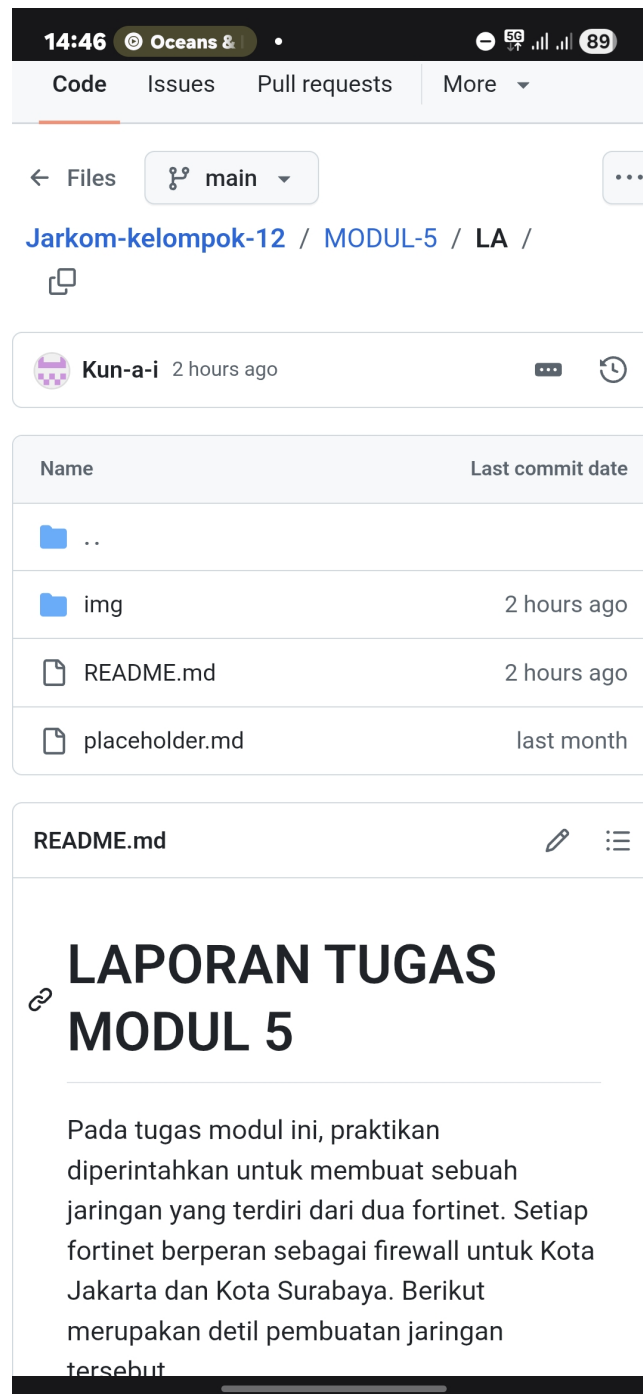
Pada Praktikum 3, konfigurasi IP address antarrouter berhasil dilakukan pada tiga perangkat yang berbeda vendor, yaitu Cisco, Huawei, dan MikroTik. Seluruh link antarrouter berhasil aktif dan dapat saling ping, membuktikan bahwa konfigurasi IP antarperangkat lintas vendor berjalan dengan benar.

Pada Praktikum 4, OSPF berhasil dikonfigurasi, dimana Cisco menggunakan router-id 1.1.1.1, Huawei menggunakan 2.2.2.2, dan MikroTik menggunakan 3.3.3.3. OSPF cost diatur agar dua link Cisco <-> Huawei menjadi jalur utama dan jalur via MikroTik berfungsi sebagai backup. Seluruh router berhasil membentuk OSPF neighbor dalam status Full dan tabel routing OSPF pada masing-masing perangkat sudah terdapat semua network yang diiklankan. Pengujian end-to-end menunjukkan PC pada semua VLAN Cisco dapat ping ke PC LAN Huawei, begitu pula sebaliknya.

Pada Praktikum 5, mekanisme failover OSPF diuji secara bertahap. Saat kedua link utama Cisco <-> Huawei aktif, hasil trace route menunjukkan bahwa traffic data melewati jalur Cisco → Huawei. Ketika salah satu link dimatikan, OSPF secara otomatis mengalihkan traffic ke link utama yang tersisa tanpa melewati MikroTik. Ketika kedua link utama dimatikan, OSPF melakukan failover ke jalur backup melalui MikroTik (Cisco → MikroTik → Huawei). Setelah kedua link diaktifkan kembali, routing kembali menggunakan jalur utama secara otomatis.

3 Hasil Tugas Modul

terlampir di readme github



4 Kesimpulan

Praktikum ini menunjukkan bahwa jaringan multivendor dapat diintegrasikan menggunakan VLAN, ROAS, DHCP Server, dan routing dinamis OSPF. Dengan melakukan konfigurasi OSPF cost, kita dapat menentukan jalur utama dan backup secara otomatis, sehingga dapat dilihat pada hasil praktikum ketika jalur utama Cisco <-> Huawei terputus, traffic berpindah ke jalur backup melalui MikroTik, dan kembali ke jalur utama begitu koneksi dipulihkan. Hal ini membuktikan bahwa OSPF mampu menjaga kelancaran lalu lintas jaringan tanpa perlu pengaturan manual.

5 Lampiran

5.1 Dokumentasi saat praktikum